Flood Susceptibility Mapping Using Artificial Neural Network (ANN) in Noakhali

Abstract

This study aims to develop a flood susceptibility map using an Artificial Neural Network (ANN) model integrated with geospatial technologies. Ten flood conditioning factors—such as elevation, slope, TWI, LULC, NDVI, drainage density, and distance to rivers—were prepared using remote sensing and GIS techniques. A flood inventory was generated from historical flood data to train and validate the ANN model. After pre-processing, feature normalization, and model training with 15,000 data points, the model achieved an AUC score of 0.82 and an overall accuracy of 74%, indicating reliable performance. The final susceptibility map classified Noakhali into four zones—Very High, High, Moderate, and Low—highlighting the char lands and southern coastal belts as the most flood-prone areas.

Introduction

Flooding is one of the most frequent and destructive natural hazards in Bangladesh, especially in low-lying coastal regions like Noakhali. The district is highly vulnerable due to its flat terrain, tidal influence, poor drainage, and intense monsoon rainfall. Traditional methods often struggle to capture the complex spatial factors influencing flood risks. With the advancement of geospatial technologies, integrating Remote Sensing, GIS, and machine learning—particularly Artificial Neural Networks (ANN)—offers a more accurate and data-driven approach. This study aims to generate a high-resolution flood susceptibility map for Noakhali to support disaster preparedness and sustainable land use planning.

Aim & Objective

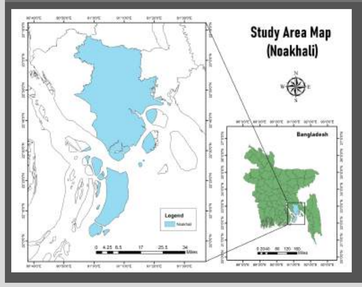
Aim:

- To develop a flood susceptibility map for Noakhali using remote sensing, GIS, and an Artificial Neural Network (ANN) model.

Objectives:

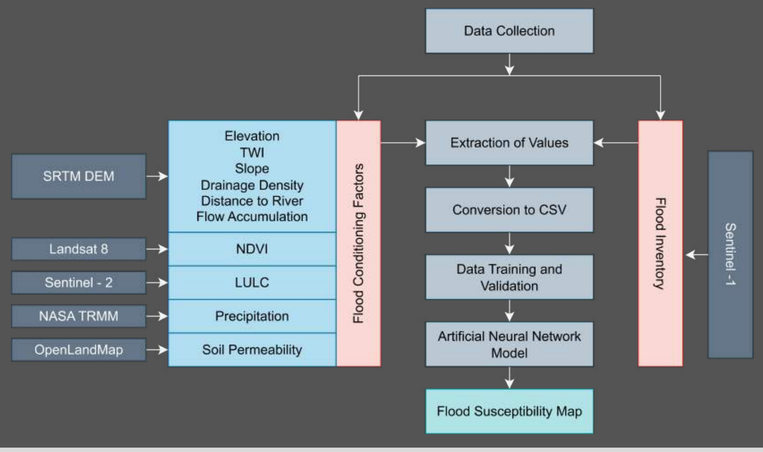
- To prepare and process flood-conditioning factors using satellite imagery and GIS techniques.
- To train and validate an ANN model based on flood inventory data to identify flood-prone areas.

Study Area

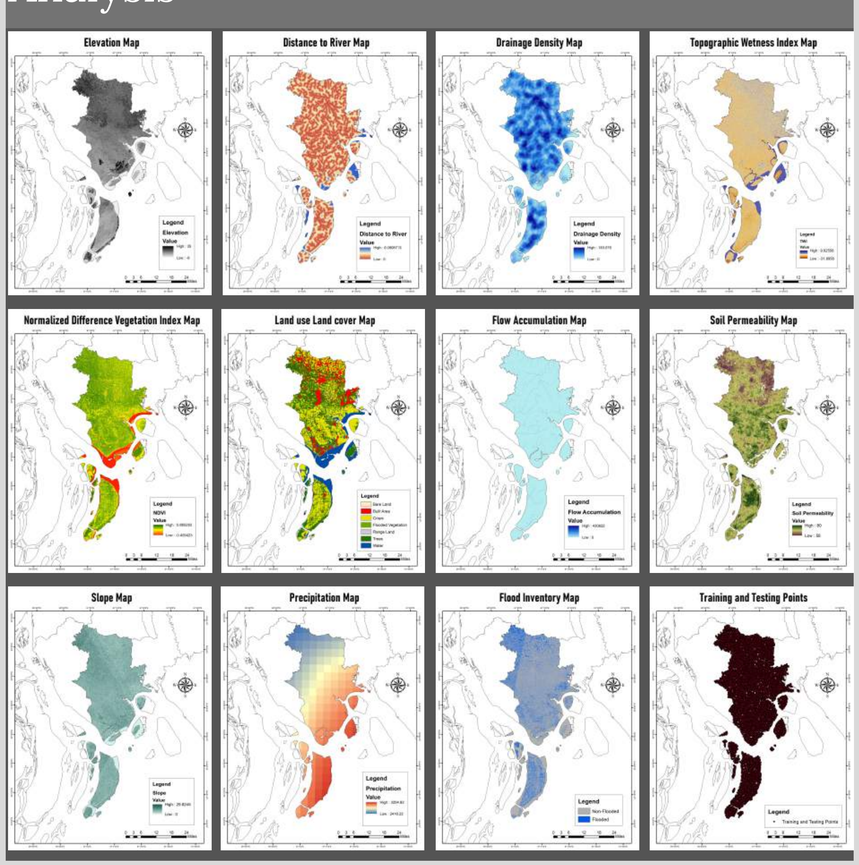


Noakhali is a coastal district in southeastern Bangladesh, characterized by flat terrain, river networks, and proximity to the Bay of Bengal. Its low elevation and poor drainage make it highly vulnerable to tidal and monsoon-induced flooding. The region includes both mainland and char areas, with land use mainly focused on agriculture. These geographical and hydrological features make Noakhali a suitable area for flood susceptibility analysis.

Methodology

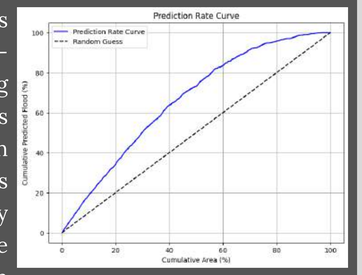


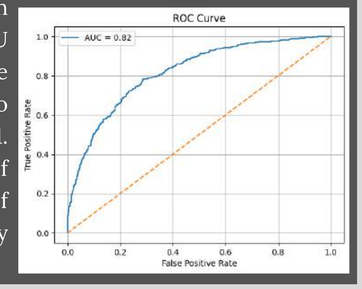
Analysis



Model Overview

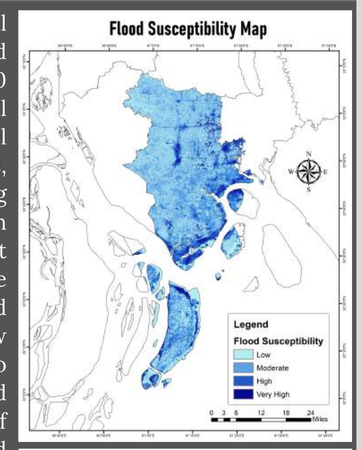
An Artificial Neural Network (ANN) was used to classify flood-prone and non-flood-prone areas based on 10 conditioning factors and 1 inventory data. The model was trained on 15,000 sample points using an 80:20 train-test split. To address class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) was applied. The ANN architecture consisted of three hidden layers (128, 64, 32 neurons) with ReLU activation and dropout regularization. The output layer used a sigmoid function to predict flood probability between 0 and 1. Model evaluation showed an accuracy of 74%, F1-score of 0.74, and an AUC score of 0.82, indicating strong classification ability and reliability for spatial flood prediction.





Findings

This study applied an Artificial Neural Network (ANN) to generate a flood susceptibility map for Noakhali using 10 flood conditioning factors and historical flood inventory data. The model successfully identified flood-prone areas, with the final classified output showing that 4.94% of the district is at very high susceptibility, 14.33% at high, 27.88% at moderate, and 52.85% at low flood risk. The southern coastal and char areas appeared most vulnerable, primarily due to low elevation, poor drainage, and proximity to rivers. The ANN model demonstrated strong performance with an AUC score of 0.82, indicating reliable classification and spatial prediction capability. Moreover, the methodology can be adapted and tested with other machine learning models like Random Forest or SVM in different flood-prone regions of Bangladesh. The findings also highlight the importance of integrating geospatial data and machine learning in addressing climate-related hazards such as flooding.



Class	Percentage
Very High	4.94
High	**14.33**
Moderate	**27.88**
Low	52.85
Total	100

Submitted by (ID): 2105051
Course ID – URP 366
Course Name – Advanced GIS Lab

Submitted to:
Debasish Roy Raja (Associate Professor, DURP, CUET)
Sourav Das (Assistant Professor, DURP, CUET)

DEPARTMENT OF URBAN AND REGIONAL PLANNING
CHITTAGONG UNIVERSITY OF ENGINEERING & TECHNOLOGY
CHATTOGRAM – 4349, BANGLADESH